

Capacitance - 1

Chapter 25, R. H. and Walker 10th
edition

Capacitance

The Capacitor: A device in which electrical energy can be stored / can store.

Example: ① Batteries in camera store energy in the photoflash unit by charging a capacitor.

② The batteries can supply energy at only a modest rate, too slowly for the photoflash unit to emit a flash of light.

③ However once the capacitor is charged, it can supply energy at a greater rate.

④ The physics of capacitors can be generalised to other devices and to any situation involving electric fields.

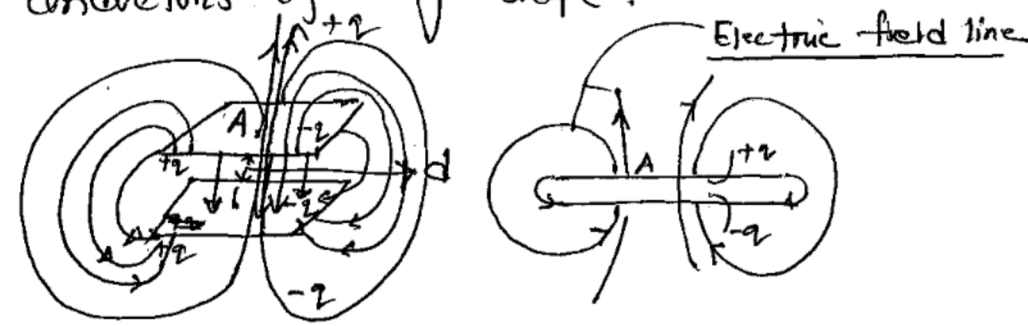
→ Earth electric field is modeled by meteorologists — as the form of spherical capacitor — partially discharged by lightning.

⑤ The charge that ice skis collect can be modeled as being stored in a capacitor that frequently discharged as sparks.

— Our discussion is — how much charge a capacitor can store. how much is denoted as Capacitance.

Capacitance Basic elements of any Capacitor —

two isolated conductors of any shape.



* Symbol of capacitor (—|—|—)

□ here $A \rightarrow$ Plate surface area

* When a capacitor is charged, its plates have charges of equal magnitudes but opposite signs: $+q$ and $-q$.

[Note: Net charge of the capacitor is zero.]

□ The capacitor has equipotential surfaces — due to? $\rightarrow$

(Plates are conductors and charge can move to reach its equilibrium state) $\rightarrow$ surface is equipotential all points on the surface are on same potential.

But there is potential difference between the two plates
↓

The charge q and the potential difference V for a capacitor are proportional to each other, that is,

$$q = CV \quad (V \rightarrow \text{potential difference, not the potential in each surface}).$$

$\rightarrow$ proportionality constant 'C' is called the capacitor capacitance of the respected capacitor.

□ Capacitance depends only on the geometry of the plates and not on their charge or potential difference.

But, the greater the capacitance, the more charge is required

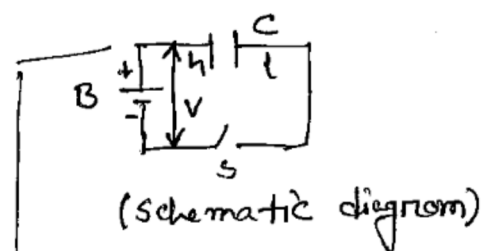
Unit: $1 \text{ farad} = 1F = 1 \text{ Coulomb per volt} = 1C/V$.

Charging a Capacitor

A battery: A battery is a device that maintains a certain potential difference between its terminals, by means of internal electrochemical reactions - resultant electric force can move internal charge.

→ Battery can charge capacitor

→ Electrons are driven through the wires by an electric field that the battery sets up in the wires.



→ Positive terminal / higher potential terminal

→ The field drives electrons from capacitor plates to the positive terminal of the battery; thus, plate h, losing electrons, becomes positively charged.

→ Ee. field drives as many electrons from the negative terminal of the battery to capacitor plate l, becomes negatively charged just as much as plate h, losing electron.

$\begin{matrix} + \\ \text{---} \\ - \end{matrix}$
Battery

$\begin{matrix} + \\ \text{---} \\ - \end{matrix}$
Capacitor

Q. What would be the capacitor potential difference if we connect the capacitor in circuit with battery?

Q. When there is no electric field to drive electron in battery, - capacitor circuit?

Checkpoint 1 - ?

Calculating the Capacitance $\rightarrow C = \frac{q}{V}$ - Total Charge
 $V \rightarrow$ Potential difference.

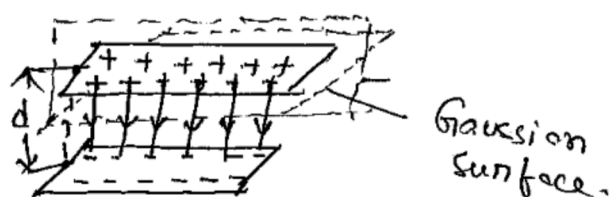
Procedure

Parallel plate capacitor

(1) Assume a charge q on the plates,

(2) Calculate the electric field $\vec{E}$ between the plates in terms of this charge (? Gauss law).

(3) If we know $\vec{E}$, we can calculate V , and finally $C = \frac{q}{V}$.



Calculating the electric field

To relate the electric field $\vec{E}$ between the plates of a capacitor to the charge q on either plate, we use Gauss' law;

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$

$\downarrow$
 electric flux.

q is the charge enclosed by a Gaussian surface

Considering E is uniform and vector $\vec{E}$ and $d\vec{A}$ is parallel

$$q = \epsilon_0 EA$$

Gaussian surface in such a way that it completely encloses the charge on the positive plate.



Calculating the potential Difference

The potential difference between the plates of a Capacitor is related to the field $\vec{E}$ by

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$$

In which the integral is to be evaluated along any path that starts on one place and ends on the other.

→ We will always choose a path that follows an electric field line, from negative plate to the positive plate. Then $\vec{E}$ and $d\vec{s}$ will have opposite directions.

$$\vec{E} \cdot d\vec{s} = -E ds \quad (\text{special case})$$

$$V = \int_{-}^{+} E ds$$

for parallel plate Capacitor.

$$q = \epsilon_0 EA$$

$$V = Ed \quad [d \rightarrow \text{plate separation}]$$

$$C = \frac{q}{V}$$

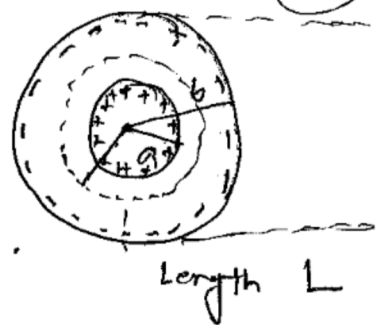
$$= \frac{\epsilon_0 EA}{Ed}$$

$$C = \frac{\epsilon_0 A}{d}$$

Thus, the Capacitance only depend on the Capacitor's geometry. for convenience $\rightarrow \epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
 $= 8.85 \text{ pF/m}.$

A cylindrical Capacitor

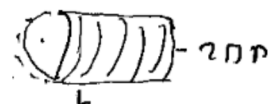
A cylindrical capacitor of length L formed by two coaxial cylinders of radii a and b . We assume that $L \gg b$ so that



we can neglect the fringing of the electric field that occurs at the ends of the cylinders. Each plate contains a charge of magnitude Q .

for Gaussian surface, we choose a cylinder of length L and radius r closed by end caps.

$$Q = \epsilon_0 E A = \epsilon_0 E (2\pi r L)$$



There is no flux through the end caps $\rightarrow$

$$E = \frac{Q}{2\pi \epsilon_0 L r} \quad \text{--- (1)}$$

$$V = \int_{-}^{+} E ds = - \frac{Q}{2\pi \epsilon_0 L} \int_b^a \frac{dr}{r} = \frac{Q}{2\pi \epsilon_0 L} \ln \left(\frac{b}{a} \right)$$

$$|ds = -dr|$$

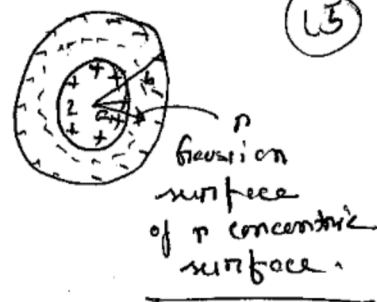
from the relation $C = \frac{Q}{V}$

(we integrate radially in word)

$$C = 2\pi \epsilon_0 \frac{L}{\ln(b/a)} \quad (\text{Cylindrical Capacitor})$$

A Spherical Capacitor

(13)



$$q = \epsilon_0 E A$$

$$q = \epsilon_0 E (4\pi r^2)$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad \text{--- (1)}$$

$$V = \int_{-1}^{+} E ds = - \frac{q}{4\pi\epsilon_0} \int_b^a \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$V = \frac{q}{4\pi\epsilon_0} \frac{b-a}{ab} \quad \text{--- (2)}$$

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}$$

Spherical Capacitor.

An isolated Sphere

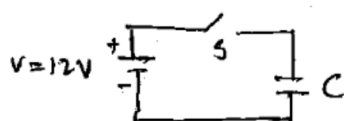
One sphere radius is tends to

infinite.

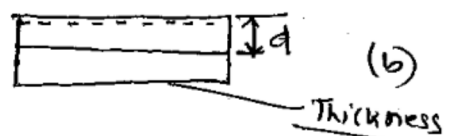
$$C = 4\pi\epsilon_0 \frac{a}{1-a/b}$$

$$b \rightarrow \infty \quad C = 4\pi\epsilon_0 R \quad (\text{isolated sphere}).$$

25.01



(a)



$$C = 0.25 \mu\text{F} \quad L = 0.50 \text{ cm}$$

$$A = 2.0 \times 10^{-4} \text{ m}^2$$

Capacitor made with copper, $n = 8.49 \times 10^{28}$ electrons/ m^3 (conduction electron)

From what depth d , within the plate, must electrons move to the plate face as the capacitor becomes charged?

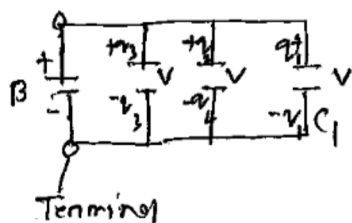
$$q = CV = (0.25 \times 10^{-6} \text{ F}) (12 \text{ V}) = 3.0 \times 10^{-6} \text{ C}$$

$$N = \frac{q}{e} = 1.873 \times 10^{13} \text{ electron (within Capacitor)}$$

$$\text{or } d = \frac{N}{An} = \frac{1.873 \times 10^{13} \text{ electrons}}{(2.0 \times 10^{-4} \text{ m}^2) (8.49 \times 10^{28} \text{ electrons/m}^3)} = 1.1 \text{ pm}$$

Note: We commonly say that electrons move from the battery to the negative face, but, actually, the battery sets up an electric field in the wires and plate such that electrons very close to the plate free move up to the negative face.

Capacitor in Parallel and in Series



① When a potential difference V is applied across several capacitors connected in parallel, that potential difference V is applied across each capacitor. The total charge q stored on the capacitors is the sum of the charges stored on all the capacitors.

→ The above three capacitors can be replaced with one capacitor.

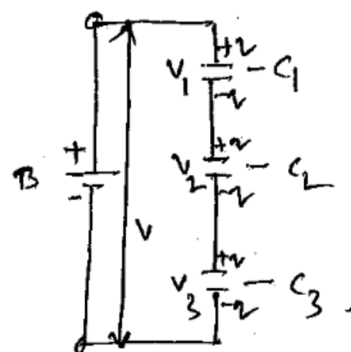
$$q_1 = C_1 V, \quad q_2 = C_2 V, \quad q_3 = C_3 V$$

$$q = q_1 + q_2 + q_3$$

$$= (C_1 + C_2 + C_3) V$$

$$C_{eq} = \frac{q}{V} = (C_1 + C_2 + C_3)$$

$$= \sum_{j=1}^n C_j \quad (\text{n capacitors in parallel})$$



② When a potential difference V is applied across several capacitors connected in series, the capacitors have identical charge q . The sum of the potential differences across all the capacitors is equal to the applied potential difference V .

$$V_1 = \frac{q}{C_1}, \quad V_2 = \frac{q}{C_2} \quad \text{and} \quad V_3 = \frac{q}{C_3}$$

$$V = V_1 + V_2 + V_3 = q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$C_{eq} = \frac{q}{V} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}} \Rightarrow \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \Rightarrow \frac{1}{C_{eq}} = \sum_{j=1}^n \frac{1}{C_j}$$

Sample problem 25.02

Capacitors in parallel and series.

Find the equivalent capacitance for the combination of capacitance, where $C_1 = 12.0 \mu F$, $C_2 = 5.30 \mu F$, and $C_3 = 4.50 \mu F$

